

Flipkart Analytics Dashboard

End-to-End E-Commerce Product Analysis

A comprehensive data engineering and business intelligence project transforming raw retail data into actionable insights. Powered by **Python** for data cleaning, **SQL Server** for deep query analysis, and **Power BI** for interactive visualization.

</> PYTHON

SQL SERVER

POWER BI





The Business Challenge

E-commerce platforms generate massive volumes of structured product data. Understanding this data is critical for optimizing pricing, discount strategies, and inventory mix.



Pricing Strategy

Identifying high-value premium products and category-wise price benchmarks.



Discount Optimization

Analyzing discount behavior to maximize customer savings and platform volume.



Outlier Detection

Isolating unusually high-priced luxury items for targeted marketing and analysis.

Dataset Profile & Architecture

The project utilizes a structured retail dataset sourced from Kaggle, refined to support robust business intelligence reporting.

Dataset Metrics

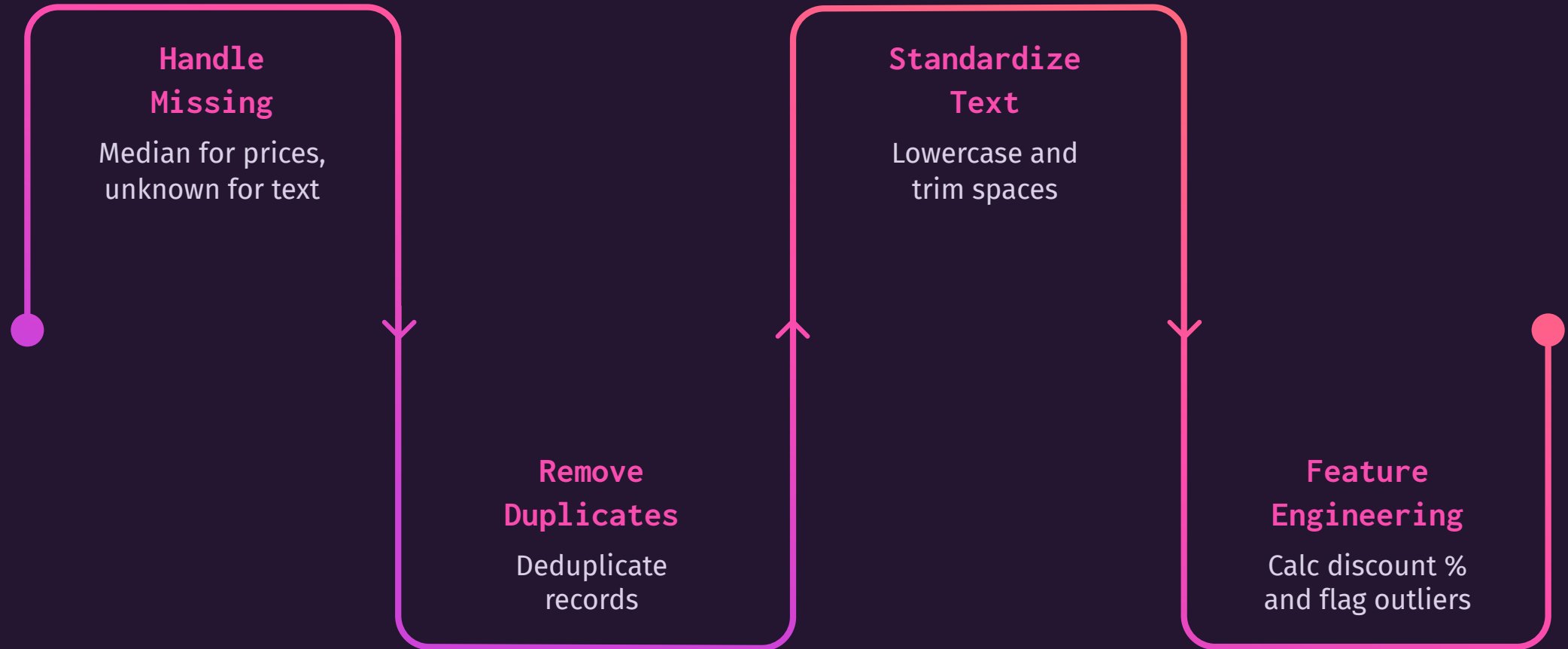
- **Source:** Kaggle Flipkart Sample
- **Cleaned Records:** 20,001 products
- **Attributes:** 17 final columns
- **Target Scope:** Prices, ratings, brands, and categories

Core Data Dictionary

uniq_id	Unique product identifier
retail_price	Original listed product price
discounted_price	Selling price after discount
discount_percent age	Feature engineered discount metric
price_outlier	Flag identifying extreme price points

Data Cleaning Pipeline

Using Python and Pandas, the raw, noisy e-commerce data was systematically cleaned and prepared for database ingestion.



- ✅ **Outlier Detection:** Used the Interquartile Range (IQR) method on retail prices to successfully identify and flag 2,063 premium outlier products.

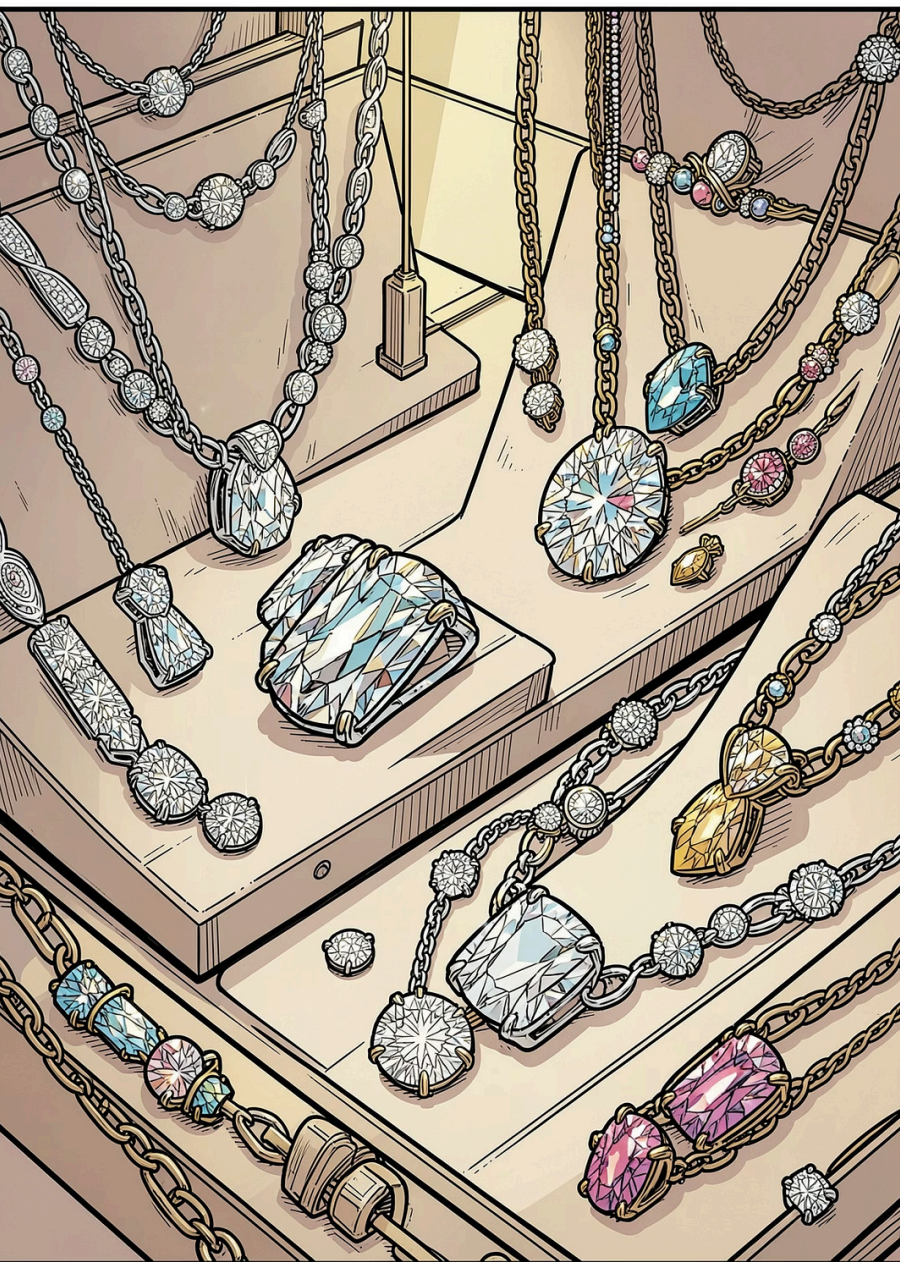
SQL Analysis: Pricing & Savings

Cleaned data was loaded into SQL Server to execute business queries. The analysis revealed a significant gap between retail and actual selling prices.



Key Pricing Insights

- **Average Retail Price:** ~INR 2.97K
- **Average Discounted Price:** ~INR 1.97K
- **Premium Outliers:** Breitling Analog Watch topped the list at a retail price of INR 571,230.
- **Maximum Savings:** Premium furniture (Durian Sofas) offered the highest absolute savings, reaching up to INR 87,675 off retail price.



Category & Brand Dominance

SQL queries identified which product categories and brands dominate the platform's listings.

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Top Product Categories

Jewellery dominates the listings, led by Necklaces (1,567 products) and Rings (710 products), followed closely by Automotive Accessories (522 products).

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Brand Distribution

A large portion of products have unbranded listings (5,865 unknown). Among established brands, *Allure Auto* (469) and *Voylla* (299) lead in product count.

Discount Strategy Analysis

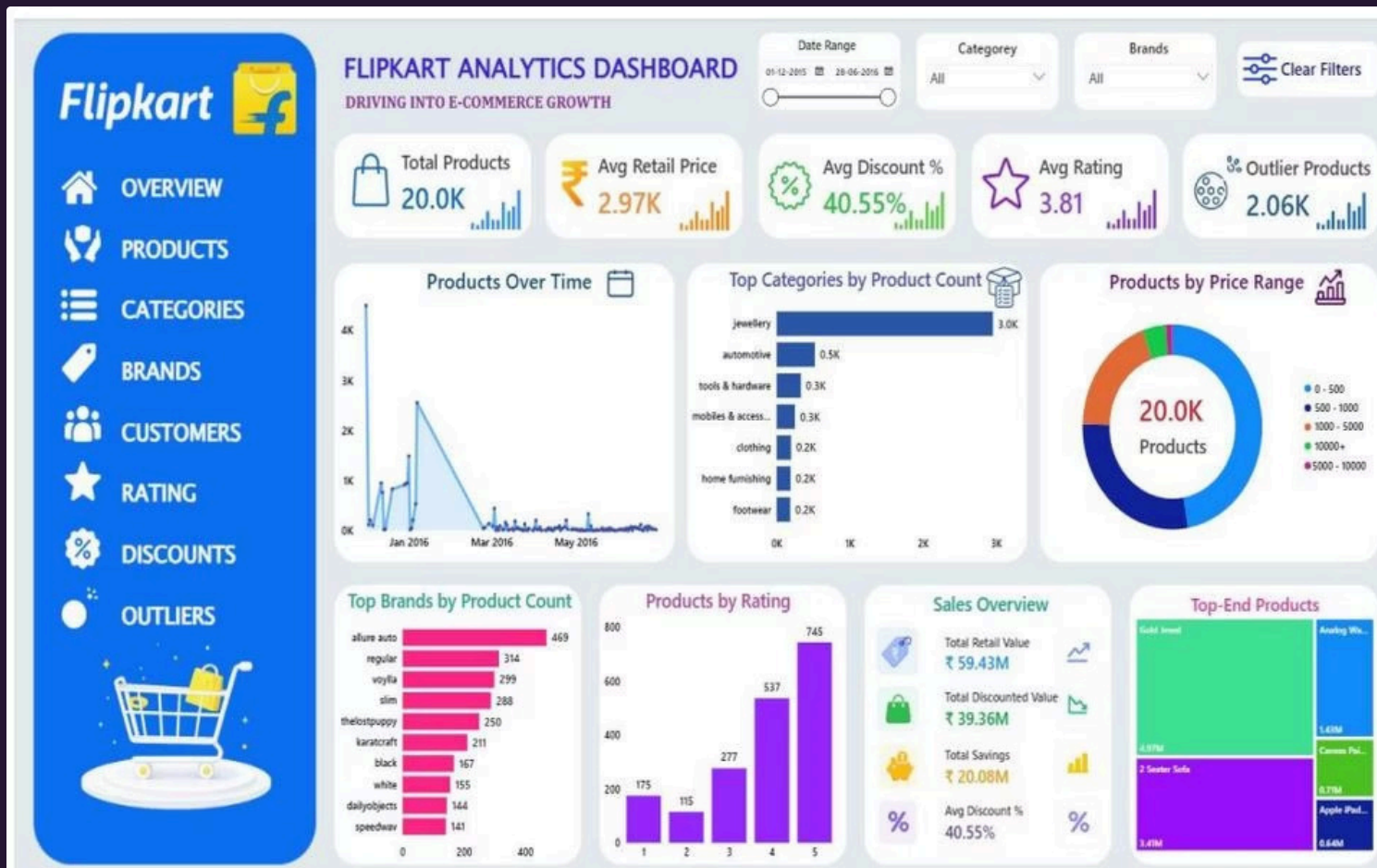
E-commerce platforms rely heavily on promotional pricing. Our SQL analysis mapped out the most aggressive discount structures.

Top Discounted Categories

Category Tree	Avg Discount %
Mobiles & Accessories >> Tablet Accessories >> Cases	94.55%
Beauty & Personal Care >> Makeup >> Vanity Boxes	91.95%
Computers >> Laptop Accessories >> USB Gadgets	91.72%

Extreme Discounts

Certain products like the *Rajcrafts Floral Double Quilt* offered discounts as high as **96.53%**, dropping from INR 6,000 to just INR 208.



Interactive Power BI Dashboard

The final phase of the project translated SQL insights into a modern, interactive executive dashboard.

KPI Cards

Instant visibility into Total Products (20K), Avg Discount (40.55%), and Outlier Products (2.06K).

Interactive Filters

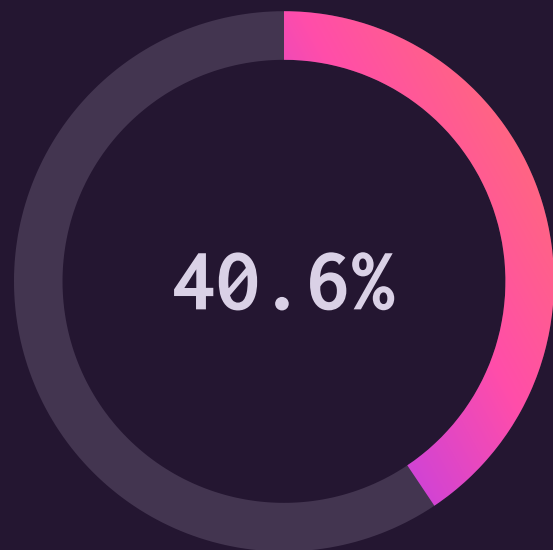
Dynamic slicing by Date Range, Category, and Brand with a single-click clear filter button.

Visual Distribution

Donut charts for price ranges, bar charts for rating distributions, and treemaps for luxury items.

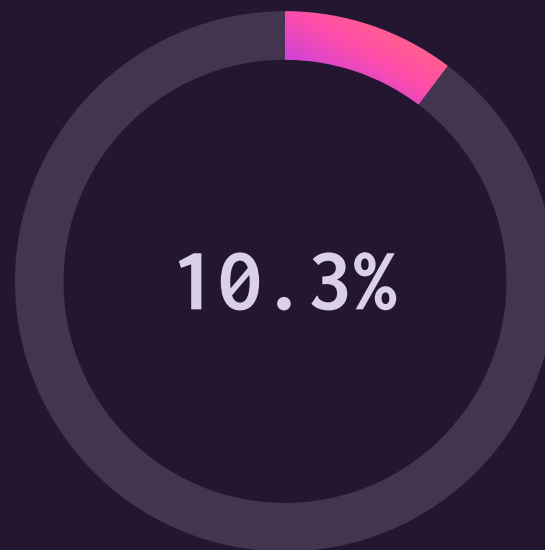
Actionable Business Insights

The analytics workflow uncovered critical patterns in Flipkart's product catalog that can drive strategic decisions.



Average Discount

High baseline discounts across the platform indicate aggressive price competition.



Price Outliers

2,063 products flagged as premium outliers, representing high-margin luxury segments.



Average Rating

Overall customer satisfaction score calculated after professionally filtering out unrated products.

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